Database Management System (DBMS)

Important Notes • Simple & Exam-Friendly •

1. Introduction to DBMS

Database: A database is an organized collection of related data that can be easily stored, accessed, managed, and updated.

DBMS: Database Management System is software used to create, store, organize, retrieve, update, and manage data in a database.

Examples: MySQL, Oracle, PostgreSQL, Microsoft SQL Server, SQLite.

2. Advantages of DBMS

- Reduces data redundancy (duplicate data).
- Provides better data security.
- Maintains data consistency.
- Supports data sharing among multiple users.
- Provides backup and recovery facilities.
- Makes data searching and updating easier.
- Provides data integrity and controlled access.

3. DBMS vs File System

Feature	File System	DBMS
Data Redundancy	High	Low
Security	Limited	Better
Data Sharing	Difficult	Easy
Backup & Recovery	Limited	Available
Data Consistency	Difficult	Better
Concurrent Access	Limited	Supported

4. Types of DBMS

- **Hierarchical DBMS:** Data is organized in a tree-like structure.
- **Network DBMS:** Data is organized as records connected through relationships.
- **Relational DBMS (RDBMS):** Data is stored in tables containing rows and columns.
- **Object-Oriented DBMS:** Data is represented as objects, similar to object-oriented programming.

5. Relational Database Concepts

- **Table:** Collection of related data arranged in rows and columns.
- **Row / Tuple:** A single record in a table.
- **Column / Attribute:** A property or field of a table.
- **Domain:** Set of valid values allowed for an attribute.
- **Schema:** Overall logical structure/design of a database.

6. Keys in DBMS

Key	Meaning
Primary Key	Uniquely identifies each record; cannot contain NULL values.
Foreign Key	Connects one table to another using a referenced key.
Candidate Key	Attribute or set of attributes that can uniquely identify a record.
Alternate Key	Candidate key not selected as the primary key.
Composite Key	Key made using two or more attributes.

7. SQL (Structured Query Language)

SQL is a standard language used to create, access, modify, and manage data in relational databases.

DDL – Data Definition Language

Used to define or change database structure.

Commands: CREATE, ALTER, DROP, TRUNCATE

DML – Data Manipulation Language

Used to insert, modify, and delete data.

Commands: INSERT, UPDATE, DELETE

DQL – Data Query Language

Used to retrieve data.

Command: SELECT

DCL – Data Control Language

Used to control user permissions.

Commands: GRANT, REVOKE

TCL – Transaction Control Language

Used to manage database transactions.

Commands: COMMIT, ROLLBACK, SAVEPOINT

8. Important SQL Examples

- **Create table:** CREATE TABLE Student (id INTEGER PRIMARY KEY, name VARCHAR(50));
- **Insert record:** INSERT INTO Student VALUES (1, 'Anu');
- **Display records:** SELECT * FROM Student;
- **Update record:** UPDATE Student SET name='Asha' WHERE id=1;
- **Delete record:** DELETE FROM Student WHERE id=1;
- **Filter records:** SELECT * FROM Student WHERE id=1;

9. Constraints

- **NOT NULL:** Prevents NULL values.
- **UNIQUE:** Ensures values are not duplicated.
- **PRIMARY KEY:** Uniquely identifies records.
- **FOREIGN KEY:** Maintains relationship between tables.
- **CHECK:** Restricts values according to a condition.
- **DEFAULT:** Provides a default value when no value is supplied.

10. Normalization

Normalization is the process of organizing data to reduce redundancy and improve data consistency.

- **1NF:** Each field contains atomic (single) values and repeating groups are removed.
- **2NF:** Table is in 1NF and has no partial dependency on a composite key.
- **3NF:** Table is in 2NF and has no transitive dependency.

11. Relationships

- **One-to-One (1:1):** One record in table A is related to one record in table B.
- **One-to-Many (1:M):** One record in table A can relate to many records in table B.
- **Many-to-Many (M:M):** Many records in A can relate to many records in B.

12. Transactions & ACID Properties

- **Transaction:** A sequence of database operations treated as one logical unit of work.
- **Atomicity:** Transaction happens completely or not at all.
- **Consistency:** Database remains valid before and after a transaction.
- **Isolation:** Concurrent transactions do not improperly affect each other.
- **Durability:** Committed changes remain saved even after a failure.

13. Joins

- **INNER JOIN:** Returns matching records from both tables.
- **LEFT JOIN:** Returns all records from the left table and matching records from the right table.
- **RIGHT JOIN:** Returns all records from the right table and matching records from the left table.
- **FULL OUTER JOIN:** Returns matching and non-matching records from both tables.

14. Index

An index is a database object that helps retrieve records faster. It can improve search performance, but indexes require additional storage and can increase the cost of insert/update operations.

15. DBMS Important Exam Points

- DBMS manages and controls access to databases.
- RDBMS stores data in related tables.
- Primary Key uniquely identifies a record.
- Foreign Key creates a relationship between tables.
- SQL is used to work with relational databases.
- Normalization reduces redundancy.
- ACID properties ensure reliable transactions.
- Constraints help maintain data integrity.

Quick Revision

Term	Remember
DBMS	Software to manage databases
RDBMS	Data stored in related tables
Primary Key	Unique identification
Foreign Key	Table relationship
SQL	Language for relational databases

Term	Remember
Normalization	Reduces redundancy
ACID	Reliable transactions